Ch 6.2: Shrinkage (Part 1)

Lecture 14 - CMSE 381

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Announcements

Last time:

- Subset selection lab

Announcements:

- No homework this week
- Weds 3/2: Lecture to finish 6.2, then ask questions
- Fri 3/4: Midterm
  - In person
  - Difficulty on par with homeworks
  - Cheat sheet: One 8.5x11 paper.
  - Handwritten
  - No photocopies

New content

- Ridge regression
- The lasso

Section 1

Last time

Best subset selection

$X_1, \ldots, X_p \quad Y$

Algorithm 6.1 *Best subset selection*

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 1, 2, \ldots p$:
   (a) Fit all $\binom{p}{k}$ models that contain exactly k predictors.
   (b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. Here *best* is defined as having the smallest RSS, or equivalently largest R^2.
   ↳ training
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.
   ↳ test score

$\mathcal{M}_0 \quad \mathcal{M}_1 \quad \mathcal{M}_2 \quad \mathcal{M}_3 \quad \ldots \quad \mathcal{M}_p$

- Check 2^p models
- Step 2 checks all models based on training score
- Picks one best per model of size k, then picks among those by best test score

Forward Stepwise Selection

Algorithm 6.2 *Forward stepwise selection*

1. Let $\mathcal{M}_0$ denote the *null* model, which contains no predictors.
2. For $k = 0, \ldots, p-1$:
   (a) Consider all $p-k$ models that augment the predictors in $\mathcal{M}_k$ with one additional predictor.
   (b) Choose the *best* among these $p-k$ models, and call it $\mathcal{M}_{k+1}$. Here *best* is defined as having smallest RSS or highest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

$\mathcal{M}_0$ $\quad$ $\mathcal{M}_1$ X_{17} $\quad$ $\mathcal{M}_2$ X_1, X_{17} X_2 X_{17} $\quad$ - - - - $\quad$ $\mathcal{M}_p$

- Check $1 + p(p+1)/2$ models
- Not guaranteed to find the best model

An example for Forward Stepwise Selection

- $\emptyset$

- $\cancel{X_1}$
- X_2
- $\cancel{X_3}$
- X_4

- $X_1\ X_2$
- $\cancel{X_1\ X_3}$
- $X_1\ X_4$
- $\cancel{X_2\ X_3}$
- $X_2\ X_4$
- $\cancel{X_3\ X_4}$

- $X_1\ X_2\ X_3$
- $X_1\ X_2\ X_4$
- $X_1\ X_3\ X_4$
- $X_2\ X_3\ X_4$

- $X_1\ X_2\ X_3\ X_4$

Backward stepwise selection

Algorithm 6.3 *Backward stepwise selection*

1. Let $\mathcal{M}_p$ denote the *full* model, which contains all p predictors.
2. For $k = p, p-1, \ldots, 1$:
   (a) Consider all k models that contain all but one of the predictors in $\mathcal{M}_k$, for a total of $k-1$ predictors.
   (b) Choose the *best* among these k models, and call it $\mathcal{M}_{k-1}$. Here *best* is defined as having smallest RSS or highest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

Ways to approximate test score

$$C_p = \frac{1}{n}(\mathrm{RSS} + 2d\hat{\sigma}^2)$$

$$\mathrm{AIC} = \frac{1}{n}(\mathrm{RSS} + 2d\hat{\sigma}^2)$$

$$\mathrm{BIC} = \frac{1}{n}(\mathrm{RSS} + \log(n)d\hat{\sigma}^2)$$

rigorous justification (statsy)

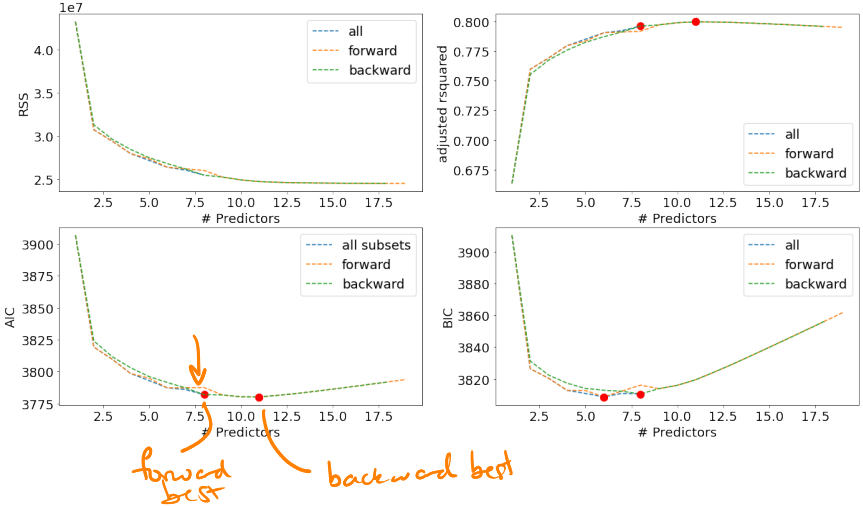
$$\text{adjusted } R^2 = 1 - \frac{\mathrm{RSS}/(n-d-1)}{\mathrm{TSS}/(n-1)}$$

intuitive, less rigorous but still useful

Cross-Validation

Previous per lecture

From Lab


1e7
RSS
Predictors
all
forward
backward
adjusted rsquared
Predictors
all
forward
backward
AIC
Predictors
all subsets
forward
backward
BIC
Predictors
all
forward
backward
forward best
backward best

Other notes from lab

- Models find different subsets
  but OK!
  B/c find similar score values

Section 2

Ridge Regression

Goal

Ridge Regression

(Lasso)

- Fit model using all p predictors
- Aim to constrain (regularize) coefficient estimates
- Shrink the coefficient estimates towards 0

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

big

small (almost 0?)

Ridge regression

Before: $y_i - \hat{y}_i$

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

Find $\beta = (\beta_0, \ldots, \beta_p)$

minimize

After:

shrin

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$

$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots$

$\lambda \geq 0$

tuning parameter / hyper parameter

$$RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$

shrinkage penalty

$\lambda = 0 \rightarrow$ least squares estimate

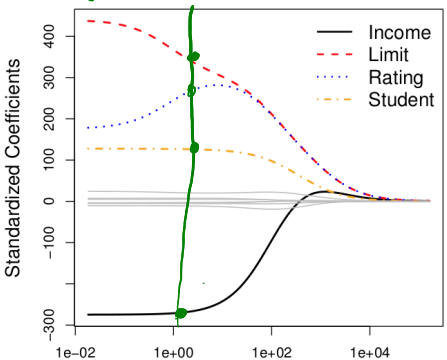
$\lambda \rightarrow \infty \rightarrow$ null model

$\hat{\beta}_\lambda^R = (\hat{\beta}_{1,\lambda}^R, \ldots, \hat{\beta}_{p,\lambda}^R)$

Example from the Credit data

$$RSS + \lambda \sum_{j=1}^{p} \beta_j^2$$

Least squares



$\lambda = 1$ λ

$\beta_0 + 300$ Limit $+ 250$ Rating $+ 110$ Student $- 280$ Income

Null model

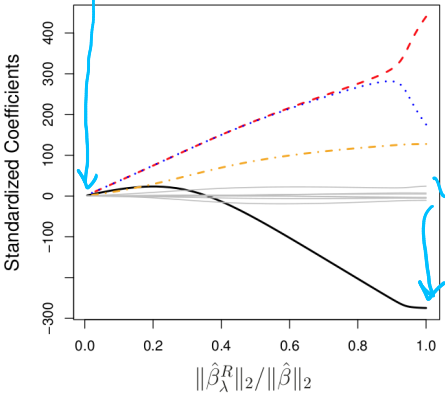
$\beta_0 + 0 \ldots$

Same Setting, Different Plot

ℓ_2 norm "ell-two"

$$RSS + \lambda \sum_{j=1}^{p} \beta_j^2 \qquad \|\beta\|_2 = \sqrt{\sum_{j=1}^{p} \beta_j^2}$$

null model



$$\underbrace{\|\hat{\beta}_\lambda^R\|_2}_{\text{leastsquares}} / \underbrace{\|\hat{\beta}\|_2}_{\text{score for least squares estimate}}$$

If $\lambda = 0 \Rightarrow \Uparrow$ is 1

If $\lambda = \text{big} \Rightarrow \Uparrow$ is ≈ 0

Scale equivavariance (or lack thereof)

Scale equivariant: Multiplying a variable by c (cX_i) just returns a coefficient multiplied by $1/c$ ($1/c\beta_i$)

Income (≈ thousands +)

$\$33,427 \; \beta_{Income}$

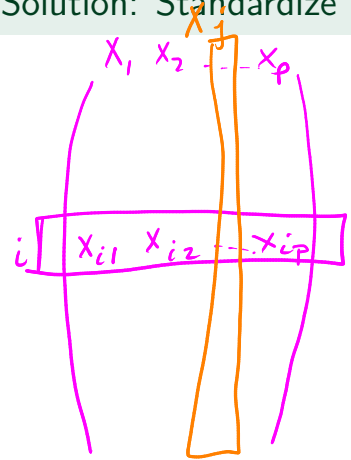
$\frac{1}{1000} \beta_{Income} \cdot 33.427$ thousands

Student (0 / 1)

Least squares is scale equivariant

$\boxed{\text{But}}$ Ridge regression is not

Solution: Standardize predictors

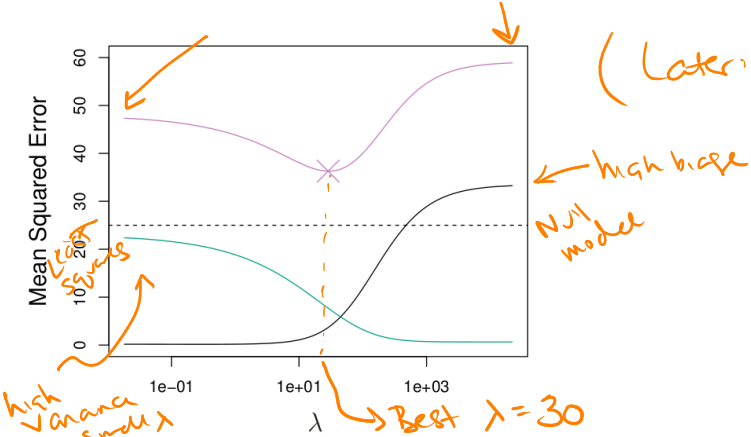


$$\widetilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \overline{x}_j)^2}}$$

} estimated std dev. of the jth predictor

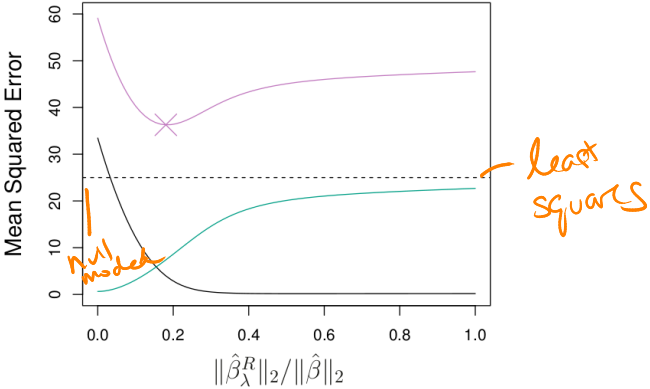
New data $\tilde{x}_{ij}$: each column has std dev 1

Bias-Variance tradeoff



Squared bias (black), variance (green), and test mean squared error (purple) for simulated data.

More Bias-Variance Tradeoff



Squared bias (black), variance (green), and test mean squared error (purple) for simulated data.

Advantages of Ridge

Ridge vs. Least Squares:

lower variance

Can use λ to trade-off

small increase in bias

for large decrease in variance

Ridge vs. Subset Selection:

Way better!

Only fit one model (per λ)

Tricks available to do all λ at once :)

Section 3

The Lasso

Same goal as before

- Fit model using all p predictors
- Aim to constrain (regularize) coefficient estimates
- Shrink the coefficient estimates towards 0

Ridge regression shrunk but did not get rid of (make 0) many coeff

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

The lasso

v1.0

Least Squares:

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

Ridge:

ℓ_2 penalty

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\underbrace{\sum_{j=1}^{p}\beta_j^2}$$

The Lasso:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = RSS + \lambda\underbrace{\sum_{j=1}^{p}|\beta_j|}$$

ℓ_1 penalty

ℓ_1 norm $\|s\|_1 = \left(\sum |s_i|\right)^{1/1}$ $\|s_2\| = \left(\sum s_i^2\right)^{1/2}$

Subsets with lasso

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = RSS + \lambda\sum_{j=1}^{p}|\beta_j|$$

- forcing some coeff to be 0
- "Subset-like" by keeping all variables w/ non-zero coeff

$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$

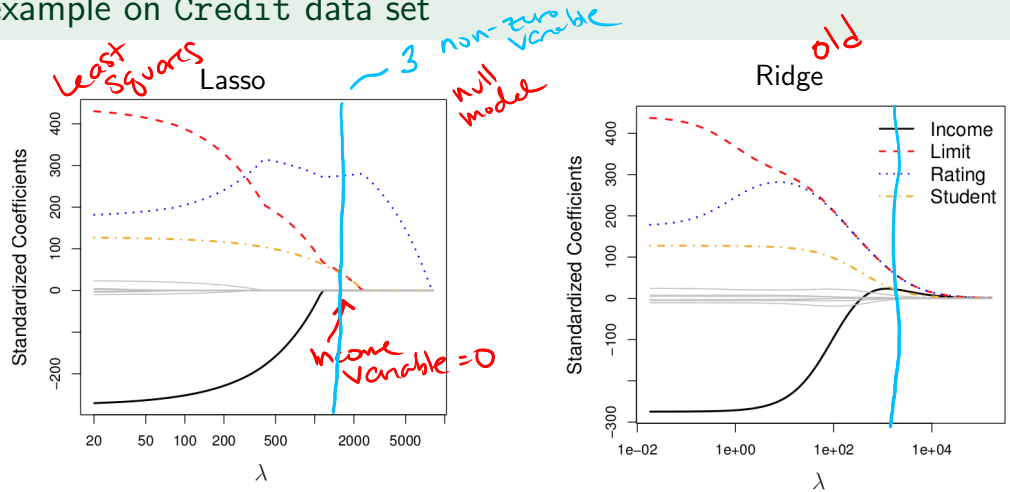
$= 27 + 0X_1 + 14X_2 - 12X_3$

- Variable selection
- Sparse models

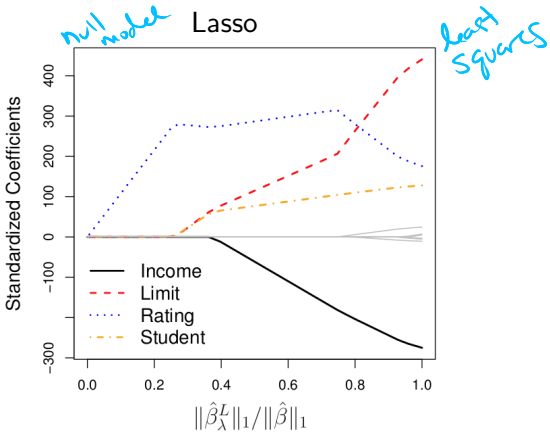
lots of 0 coeff.

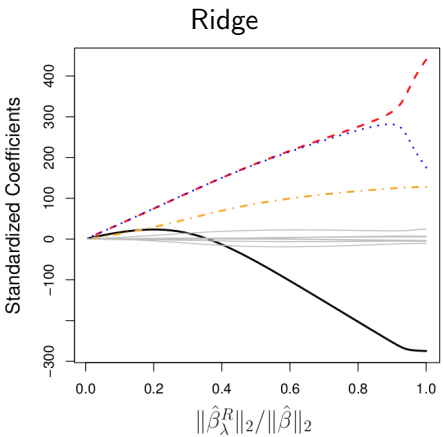
- Selecting good λ is critical!

An example on Credit data set



More example on `Credit` data set





TL;DR

Least Squares:

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

Ridge:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\underbrace{\sum_{j=1}^{p}\beta_j^2}_{\ell_2}$$

The Lasso:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = RSS + \lambda\underbrace{\sum_{j=1}^{p}|\beta_j|}_{\ell_1}$$

Next time

Next time:

- Some more properties of lasso
- Selection λ
- Questions for midterm

This week:

- Mon 2/28: 6.2 Shrinkage
- Weds 3/2: Finish shrinkage + Review session
  - Bring your questions!
- Fri 3/4: Midterm
  - In person
  - Difficulty on par with homeworks
  - Cheat sheet: One 8.5x11 paper.
  - Handwritten
  - No photocopies

A bit short,
maybe some more CV?
Maybe more lab review?